Cells Under the Microscope

Biology 2e chapter 4 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Microscopes enlarge a small field of view. An eyepiece scale has no fixed length until it is calibrated against a stage micrometer at the selected objective. Focus and illumination make boundaries easier to distinguish; magnification alone does not reveal every organelle.

Investigation question

How do the visible structures and sizes of different cells compare?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Start at 10x objective, focus, then inspect at 40x (400x total). The displayed calibration is 2 micrometers per scale division at 40x. Measure the selected cell along its long axis using the draggable ruler endpoint.

Prepare the equipment in the displayed order. Observe or classify the sample, review your measurements, then click Collect data and continue. Repeat through the assigned comparison. Explore other settings after collecting the full comparison.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: micrometers

Condition / sample	Repeat	Source	micrometers
Onion epidermis	1		
Onion epidermis	2		
Aquatic leaf	1		
Aquatic leaf	2		
Prepared animal epithelium	1		
Prepared animal epithelium	2		

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Sample / field: _____ Source: _____

Item / band / cell	Measurement or classification	Reasoning / uncertainty

For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

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Compare measured lengths and visible boundaries. Which structures can you actually resolve?

Why must the scale be recalibrated after changing objective?

How representative are two selected cells of the whole tissue?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Original schematic cells, not micrographs. Dimensions illustrate plausible ranges, not species measurements. Optical resolution, staining and specimen thickness are simplified.